

## Module 3: Periodic table and energy

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The focus of this module is inorganic and physical chemistry, the applications of energy use to everyday life and industrial processes, and current environmental concerns associated with sustainability.

The content within this module assumes knowledge and understanding of the chemical concepts developed in Module 2: Foundations in chemistry.

This module provides learners with a knowledge and understanding of the important chemical ideas that underpin the study of inorganic and physical chemistry:

- the periodic table: periodic and group properties
- enthalpy changes and their determination
- rates of reaction
- reversible reactions and chemical equilibrium
- consideration of energy and yield in improving sustainability.

This module allows learners to develop important qualitative practical skills, especially observational skills required for analysis, and accurate quantitative

techniques involved in determination of energy changes and reaction rates.

There are opportunities for developing mathematical skills when studying enthalpy changes and reaction rates and when carrying out quantitative practical work.

### Synoptic assessment

This module provides a context for synoptic assessment and the subject content links strongly with content encountered in Module 2: Foundations in chemistry.

- Atoms, moles and stoichiometry
- Acid and redox reactions
- Bonding and structure

Knowledge and understanding of Module 2 will be assumed and examination questions will be set that link its content with this module and other areas of chemistry.

### 3.1 The periodic table

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Periodic trends are first studied to extend the understanding of structure and bonding. Group properties are then studied using Group 2 and the halogens as typical metal and non-metal groups respectively, allowing an understanding of redox reactions to be developed further.

Finally, this section looks at how unknown ionic compounds can be analysed and identified using simple test-tube tests.

### 3.1.1 Periodicity

| Learning outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Additional guidance                                                                                                                                                                                                                                                                                           |
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| <i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                               |
| <b>The structure of the periodic table</b>                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                               |
| (a) the periodic table as the arrangement of elements: <ul style="list-style-type: none"><li>(i) by increasing atomic (proton) number</li><li>(ii) in periods showing repeating trends in physical and chemical properties (periodicity)</li><li>(iii) in groups having similar chemical properties</li></ul>                                                                                                                                                                           | HSW1,7,11 The development of the Periodic Law and acceptance by the scientific community.<br><br>HSW7,11 The extension of the periodic table through discovery and confirmation of new elements.                                                                                                              |
| <b>Periodic trend in electron configuration and ionisation energy</b>                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                               |
| (b) (i) the periodic trend in electron configurations across Periods 2 and 3 ( <b>see also 2.2.1 d</b> )<br>(ii) classification of elements into s-, p- and d-blocks                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                               |
| (c) first ionisation energy (removal of 1 mol of electrons from 1 mol of gaseous atoms) and successive ionisation energy, and: <ul style="list-style-type: none"><li>(i) explanation of the trend in first ionisation energies across Periods 2 and 3, and down a group, in terms of attraction, nuclear charge and atomic radius</li><li>(ii) prediction from successive ionisation energies of the number of electrons in each shell of an atom and the group of an element</li></ul> | M3.1<br><br>Definition required for first ionisation energy only. Explanation to include the small decreases as a result of s- and p-sub-shell energies (e.g. between Be and B) and p-orbital repulsion (e.g. between N and O).<br><br>HSW1,2 Trends in ionisation energy support the Bohr model of the atom. |
| <b>Periodic trend in structure and melting point</b>                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                               |
| (d) explanation of: <ul style="list-style-type: none"><li>(i) metallic bonding as strong electrostatic attraction between cations (positive ions) and delocalised electrons</li><li>(ii) a giant metallic lattice structure, e.g. all metals</li></ul>                                                                                                                                                                                                                                  | No details of cubic or hexagonal packing required.                                                                                                                                                                                                                                                            |
| (e) explanation of the solid giant covalent lattices of carbon (diamond, graphite and graphene) and silicon as networks of atoms bonded by strong covalent bonds                                                                                                                                                                                                                                                                                                                        | HSW1,9 Use of ideas about bonding to explain the strength and conductive properties of graphene, and its potential applications and benefits.                                                                                                                                                                 |

- (f) explanation of physical properties of giant metallic and giant covalent lattices, including melting and boiling points, solubility and electrical conductivity in terms of structure and bonding
- (g) explanation of the variation in melting points across Periods 2 and 3 in terms of structure and bonding (**see also 2.2.2 o**).

Explanations should be in terms of the types of particle present in a lattice, the relative strength of forces and bonds, and the mobility of the particles involved, as appropriate.

HSW1 Use of ideas about bonding to explain macroscopic properties.

*M3.1*

Trend in structure from giant metallic to giant covalent to simple molecular lattice.